

The Constants of Nature

$$\begin{array}{l}
d_{ca} = \frac{E_1}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
\text{eV} : \text{Hz} = \frac{eV}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
\text{J} : \text{Hz} = \frac{\text{joule}}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
\text{Hz} : \text{eV} = 2\pi \frac{eV}{\text{joule}} \left(\frac{t_p}{s f_p} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
\hbar = \frac{eV}{\text{joule}} \left(\frac{l_p^2 m_p}{t_p} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
h = \left(\frac{l_p^2 m_p}{t_p} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
h = 2\pi \left(\frac{l_p^2 m_p}{t_p} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
q_e = \pi \left(\frac{l_p^2 m_p}{t_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
2q_c = 2\pi \left(\frac{l_p^2 m_p}{t_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
\mu_B = \frac{1}{2} \left(\frac{l_p^2 C m_p}{t_p m_e} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
\mu_N = \frac{1}{2} \left(\frac{l_p^2 C m_p}{t_p m_n} \right) \left(1 + \sqrt{\zeta(2)} \mathcal{W}_0^2 \boxtimes \right) \\
G_0^{-1} = \frac{\pi}{\kappa_1^2} \left(\frac{l_p^2 m_p}{t_p q_p} \right) \left(1 + \sqrt{\zeta(3)} \mathcal{W}_0^2 \boxtimes \right) \\
G_0 = \frac{\kappa_1^2}{\pi} \left(\frac{t_p q_p^2}{l_p^2 m_p} \right) \left(1 - \sqrt{\zeta(3)} \mathcal{W}_0^2 \boxtimes \right) \\
\text{eV} = \kappa_1 \text{joule} \left(\frac{q_p}{C} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \boxtimes \right) \\
\text{J} : \text{eV} = \frac{eV}{\kappa_1} \left(\frac{C}{q_p} \right) \left(1 - \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \boxtimes \right) \\
-\frac{e}{m_e} = -\kappa_1 \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \boxtimes \right) \\
\frac{e}{m_e} = \kappa_1 \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \boxtimes \right) \\
\frac{\hbar}{h} = \frac{\kappa_1}{4\pi} \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \boxtimes \right) \\
\frac{\hbar_N}{h} = \frac{\kappa_1}{4\pi} \frac{1}{\text{MHz}} \left(\frac{q_p}{s m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \mathcal{W}_0^2 \boxtimes \right) \\
a = \frac{4\pi}{32 s \kappa_1^4} \left(\frac{l_p m_p}{m_e} \right) \left(1 + 35 \left(\frac{4\pi}{8} \right)^5 \mathcal{W}_0^2 \boxtimes \right) \\
d_{220} = \frac{1}{\sqrt{8}} \frac{4\pi}{32 s \kappa_1^4} \left(\frac{l_p m_p}{m_e} \right) \left(1 + 35 \left(\frac{4\pi}{8} \right)^5 \mathcal{W}_0^2 \boxtimes \right) \\
A_{90} = \text{ampere} \left(1 + \frac{4\pi}{35} \right) \mathcal{W}_0^3 \boxtimes \\
C_{90} = \text{coulomb} \left(1 + \frac{4\pi}{35} \right) \mathcal{W}_0^3 \boxtimes \\
\mu_B = \frac{\kappa_1}{4\pi} \left(\frac{t_p q_p}{l_p m_e} \right) \left(1 + s \left(\frac{K}{6} \right) \mathcal{W}_0^3 \boxtimes \right) \\
\frac{\hbar_c}{h_c} = \frac{\kappa_1}{4\pi} \left(\frac{t_p q_p}{l_p m_e} \right) \left(1 + s \left(\frac{K}{6} \right) \mathcal{W}_0^3 \boxtimes \right) \\
R_{\omega} = \frac{\kappa_1^4}{4\pi} \left(\frac{m_e}{l_p m_p} \right) \left(1 - \frac{6}{V_{fe}} \right)^2 \mathcal{W}_0^2 \boxtimes \\
E_b \frac{1}{m} = \frac{\kappa_1^4}{2\pi} \left(\frac{m_e}{l_p m_p} \right) \left(1 - \frac{6}{V_{fe}} \right)^2 \mathcal{W}_0^2 \boxtimes \\
\frac{1}{m} : E_b = \frac{2\pi}{\kappa_1^4} \left(\frac{l_p m_p}{m m_e} \right) \left(1 + \frac{6}{V_{fe}} \right)^2 \mathcal{W}_0^2 \boxtimes \\
A_{\text{time}} = \left(\frac{l_p m_p}{\kappa_1^4 m_e} \right) \left(1 + \sqrt{5} \left(\frac{C_d}{C_U} \right) \mathcal{W}_0^4 \boxtimes \right) \\
\text{Hz} : E_b = \frac{2\pi}{\kappa_1^4} \left(\frac{t_p m_p}{m_e} \right) \left(1 + \sqrt{5} \left(\frac{C_d}{C_U} \right) \mathcal{W}_0^4 \boxtimes \right) \\
E_b : \text{Hz} = \frac{\kappa_1^4}{2\pi} \left(\frac{t_p m_p}{m_e} \right) \left(1 - \sqrt{5} \left(\frac{C_d}{C_U} \right) \mathcal{W}_0^4 \boxtimes \right) \\
R_{\omega^*c} = \frac{\kappa_1^4}{4\pi} \left(\frac{m_e}{t_p m_p} \right) \left(1 - \sqrt{5} \left(\frac{C_d}{C_U} \right) \mathcal{W}_0^4 \boxtimes \right) \\
\kappa = \left(\frac{l_p^3 m_p}{t_p^3 q_p^2} \right) \left(1 + \left(\frac{C_d}{6} \right) \mathcal{W}_0^4 \boxtimes \right) \\
A_{\text{perm}} = \left(\frac{l_p^3 m_p}{t_p^3 q_p^2} \right) \left(1 - \left(\frac{C_d}{6} \right) \mathcal{W}_0^4 \boxtimes \right) \\
\varepsilon_0 = \frac{1}{4\pi} \frac{t_p^2 q_p^2}{\omega^2 s} \left(1 - \left(\frac{C_d}{6} \right) \mathcal{W}_0^4 \boxtimes \right) \\
A_{\text{mod}} = \kappa_1 \left(\frac{l_p^2 q_p m_p}{t_p m_p} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{W}_e^2 \boxtimes \right) \\
\mu_B = \frac{\kappa_1}{2} \left(\frac{l_p^2 q_p m_p}{t_p m_e} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{W}_e^2 \boxtimes \right) \\
\mu_N = \frac{\kappa_1}{2} \left(\frac{l_p^2 q_p m_p}{t_p m_n} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{W}_e^2 \boxtimes \right) \\
\mu_{\text{tri}} = \kappa_1^2 \mathcal{W}_e^2 \left(D_{\text{Do}} + \pi \right) \left(\frac{l_p^2 q_p m_p}{t_p m_{\text{tri}}} \right) \left(1 - \omega_2 \left(\frac{2\pi}{18} \right) \mathcal{W}_e^2 \boxtimes \right) \\
\frac{\mu_{\text{tri}}}{\mu_N} = S \left(D_{\text{Do}} + \pi \right) \left(\frac{m_e}{m_{\text{tri}}} \right) \left(1 + \frac{4}{L_{\text{Li}}} \left(\frac{32}{45} \right) \mathcal{W}_e^2 \boxtimes \right) \\
\frac{\mu_{\text{tri}}}{\mu_N} = \kappa_1 \mathcal{W}_e^2 \left(2D_{\text{Do}} + 2\pi \right) \left(\frac{m_e}{m_{\text{tri}}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{W}_e^2 \boxtimes \right) \\
\frac{\mu_{\text{tri}}}{\mu_B} = 2 \kappa_1 \mathcal{W}_e^2 \left(2D_{\text{Do}} + 2\pi \right) \left(\frac{m_e}{m_{\text{tri}}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{W}_e^2 \boxtimes \right) \\
g_{\text{tri}} = \kappa_1 \mathcal{W}_e^2 \left(4D_{\text{Do}} + 4\pi \right) \left(\frac{m_e}{m_{\text{tri}}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{W}_e^2 \boxtimes \right) \\
\mu_m = -\frac{S}{C_{\text{CFP}}} \left(\frac{m_e}{m_n} \right) \left(1 + 15 \left(\frac{2\pi}{8K} \right) \mathcal{W}_e^2 \boxtimes \right) \\
\frac{\mu_e}{\mu_n} = -\frac{C_{\text{CFP}}}{S} \left(\frac{m_n}{m_e} \right) \left(1 + 15 \left(\frac{2\pi}{8K} \right) \mathcal{W}_e^2 \boxtimes \right)^{-1} \\
\frac{\mu_n}{\mu_e} = \frac{C_{\text{CFP}}}{P_{\text{up}}} \left(\frac{m_n}{m_e} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{W}_e^2 \boxtimes \right) \\
\frac{\mu_e}{\mu_e} = \frac{P_{\text{up}}}{C_{\text{CFP}}} \left(\frac{m_e}{m_n} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{W}_e^2 \boxtimes \right)^{-1} \\
\mu_{\mu} = \left(\frac{m_{\mu}}{m_e} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{W}_e^2 \boxtimes \right) \\
G = \left(\frac{l_p^3}{t_p^2 m_p} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{W}_e^3 \boxtimes \right) \\
\hbar C = \text{GeV}^2 \left(\frac{t_p^4}{l_p^4 m_p^2} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{W}_e^3 \boxtimes \right) \\
m_n = \left(\frac{2\pi^2}{25} \right) \frac{\text{GeV}}{\omega^2} \left(\frac{t_p^2}{l_p^2} \right) \left(1 - \left(\frac{2s}{\omega} \right) \mathcal{W}_e^3 \boxtimes \right) \\
m_e = \left(\frac{2\pi^2}{s} \right) \frac{\text{GeV}}{\omega_2} \left(\frac{t_p^2}{l_p^2} \right) \left(1 - \frac{\sqrt{2}}{e^{\gamma/3}} \left(\frac{18}{25} \right) \mathcal{W}_e^3 \boxtimes \right) \\
m_p = \frac{2\pi}{35} \left(\frac{4\pi}{6s} \right) (A_{\text{mass}}) \left(1 - 18 K_{-6} \mathcal{W}_e^3 \boxtimes \right) \\
\mu_B = -P_{\text{up}} \left(\frac{2s}{6\pi} \left(\frac{m_e}{m_{de}} \right) \right) \left(1 -$$

$m_p = 5.39125836832313 \times 10^{-34}$ s	Planck time
$l_p = 1.61625918715645 \times 10^{-35}$ m	Planck length
$\hbar_p = 1.87545596713962 \times 10^{-18}$ C	Planck charge
$T_p = 1.41678698597095 \times 10^{-32}$ K	Planck temperature
$m_{\text{pl}} = 2.17642683817579 \times 10^{-8}$ kg	Planck mass
$\boxtimes = 0.999999199973626 \times 10^{-7}$	hyperbolic inversion boundary
$\boxtimes$	number of dimensions
$\frac{1}{7}$	break in scale symmetry
$\kappa_1 = 0.0854245431533304 \dots$	1 st hyperbolic partition constant
$\kappa_2 = 3.667567573485501 \dots$	2 nd hyperbolic partition constant
$\kappa_3 = 1.87649603900417 \dots + 0.46615262615972 \dots i$	3 rd hpc
$\kappa_4 = 1.87649603900417 \dots - 0.46615262615972 \dots i$	4 th hpc
$\kappa_5 = 4.47826244916551 \dots$	hyperbolic radius constant
$\kappa_6 = 2.00316562310924 \dots$	hyperbolic radial constant
$G_{40} = 1.01494160640965 \dots$	Gieseking's constant
$V_E = 0.029838212819390 \dots$	figure eight knot hyperbolic volume
$K = 2.0956559471727190 \dots$	Catalan's constant
$C_d = 0.291560904030818 \dots$	domino tiling constant
$D_4 = 1.79162281206959 \dots$	dimer constant
$\omega_1 = 0.7649797018528596 \dots + 1.32497062714087 \dots i$	omega_1 constant
$\omega_2 = 1.5259430730571932 \dots$	omega_2 constant
$i^{i^{i^{\dots}}}$	i power tower of i
$i = (-1)^{i/3}$	imaginary unit
$q_1 = (-\frac{1}{2})^{1/3}$	imaginary golden ratio
$\pi_{10} = 1.61803398874989 \dots$	golden ratio
$\pi_{13} = 1.9152965358979 \dots$	Archimedes' constant
$P_p = 2.29558714939263 \dots$	universal parabolic constant
$s = 5.24411510858423 \dots$	arc length of the unit lemniscate
$L = 2.62025755429219 \dots$	lemniscate constant
$L_1 = 1.31028771714605 \dots$	1 st lemniscate constant
$L_2 = 0.599070117367796 \dots$	2 nd lemniscate constant
$C_{ij} = 0.847213084793979 \dots$	ubiquitous constant
$G_{64} = 0.834626841674073 \dots$	Gauss's constant
$W_{We} = 0.479494379987920 \dots$	Weierstrass constant
$C_{R1} = 0.998136044598509 \dots$	Ramanujan's 1 st continued fraction constant
$L_{11} = 0.110001000000000 \dots$	Liouville's constant
$M = -1.74756456463318 \dots$	Madelung constant
$p = 2.718284201072035 \dots$	prime constant
$e = 2.71828182845904 \dots$	Euler's number
$x_{50} = 2.29316268471186 \dots$	1 st Foias constant
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$S = 0.822825249678847 \dots$	Sierpinski's constant
$D_{90} = 0.739085133215160 \dots$	Dotrice number
$L_1 = 0.662243419349391 \dots$	Laplace limit
$C_{EPP} = 1.199678640251773 \dots$	real fixed point of the hyperbolic cotangent
$J_{0,1} = 2.404825557695772 \dots$	1 st root of the Bessel function
$\alpha_F = 2.50290787509589 \dots$	alpha Feigenbaum constant
$\delta_F = 0.466920160910299 \dots$	delta Feigenbaum constant
$S = 3.24659760371714 \dots$	silver constant
$P = 1.32471795724474 \dots$	plastic constant
$C_Q = 0.605443657196732 \dots$	QRS constant
$K_{-6} = 1.15655237442151 \dots$	Khinchin mean of order -6
$T_0 = 2.731500000000000 \dots \times 10^2$ kelvin	freezing point of water
$p_0 = 1.00000000000000 \dots \times 10^5$ pascal	1 bar
$P_1 = 1.03125000000000 \dots \times 10^5$ pascal	standard atmospheric pressure
$E_1 = 6.09120229711386 \dots \times 10^{-24}$ joules	energy associated with $\Delta E_{\gamma s}$
$f_1 = 5.40000000000000 \dots \times 10^{-14}$ Hz	frequency of max light conversion
$\lambda_{\text{peak}} = 4.96511423174427 \dots$	peak 2 of spectral radiance/unit wavelength
$\nu_{\text{peak}} = 2.82143937212207 \dots$	peak emission of spectral flux/unit frequency
$n!$ = factorial function	
$\ln$ = derangement function	
$\log(x)$ = natural logarithm	$\text{lm} = \text{lumen} = \text{cd} \cdot \text{sr}$
$\Gamma(x)$ = gamma function	$N = \text{newton} = \text{m kg/s}^2$
$\zeta(\chi)$ = Riemann zeta function	$N = \text{noether} = \text{m}^2/\text{kg} \cdot \text{s}$
	$\text{N} = \text{ohm} = \text{m}^2/\text{kg} \cdot \text{s}^2/\text{C}^2$
units	$\text{Pa} = \text{pascal} = \text{kg/s}^2/\text{m}^2$
s = second	$S = \text{siemens} = \text{s}^2/\text{m}^2/\text{kg} \cdot \text{C}^2$
m = meter	$T = \text{tesla} = \text{kg/s}^2/\text{C}$
C = coulomb	$V = \text{volt} = \text{m}^2/\text{kg} \cdot \text{s}^2/\text{C}$
K = kelvin	$W = \text{watt} = \text{m}^2/\text{kg} \cdot \text{s}^3$
kg = kilogram	$\text{Wb} = \text{weber} = \text{m}^2/\text{kg} \cdot \text{s} \cdot \text{C}$
derived units	
$\text{MHz} = \text{megahertz} = 1.00000000000000 \dots \times 10^6$ 1/s	
$\text{fm} = \text{femtometer} = 1.00000000000000 \dots \times 10^{-15}$ m	
$\text{eV} = \text{electron-volt} = 1.00000000000000 \dots \text{eV}$	
$\text{MeV} = \text{megaelectron-volt} = 1.00000000000000 \dots \times 10^6$ eV	
$\text{GeV} = \text{gigaelectron-volt} = 1.00000000000000 \dots \times 10^9$ eV	
$\frac{r_a}{r_e} = \left(\frac{4\pi}{5}\right)^2$	$\frac{r_a}{r_e} = K$
$\frac{r_a}{r_e} = \sqrt{\frac{C_1}{C_2}}$	$\frac{r_e}{r_{\text{dr}}} = \sqrt{\log(3\pi)}$
$E_p = \frac{t_p^2}{m_p}$	
transform space	